C29PE Class Test (Week 7, October 2024)

Multiple Choice Questions (worth 1.5 marks each – please answer all questions)

1) The central economic problem of scarcity arises because:

A) Resources are abundant and easily accessed by everyone.

B) There are limited resources to satisfy unlimited human wants and needs.

C) Prices of goods and services are constantly fluctuating.

D) Governments fail to regulate the distribution of resources efficiently.

2) If the demand for electric cars increases due to rising environmental concerns, while simultaneously the technology for producing electric cars improves, what is the most likely effect on the equilibrium price and quantity of electric cars?

A) Equilibrium price will increase and equilibrium quantity will decrease.

B) Equilibrium price will decrease and equilibrium quantity will increase.

C) Equilibrium price may rise, fall or stay the same and equilibrium quantity will increase.

D) Equilibrium price will increase and equilibrium quantity may rise, fall or stay the same.

3) When the price of coffee increases, the demand for tea increases. This is an example of:

A) Positive cross-price elasticity, indicating that coffee and tea are substitutes.

B) Positive cross-price elasticity, indicating that coffee and tea are complements.

C) Negative cross-price elasticity, indicating that coffee and tea are substitutes.

D) Negative cross-price elasticity, indicating that coffee and tea are complements.

4) In many cities, governments impose rent control laws that set a maximum price landlords can charge for renting apartments. According to the economic theory, what is the primary economic consequence of such rent control?

A) An increase in the quantity of available rental housing.

B) A decrease in the demand for rental housing.

C) A shortage of rental housing, as the quantity demanded exceeds the quantity supplied at the controlled price.

D) An increase in the quality and maintenance of rental housing.

5) Sue decides to spend her Saturday working overtime instead of going hiking with friends. The opportunity cost of Sue's decision is:

A) The money she earns from working overtime.

B) The enjoyment and relaxation she would have experienced while hiking.

C) The additional work experience she gains from working overtime.

D) The cost of her hiking gear.

6) A small coffee shop hires more baristas to speed up service during peak hours. However, after a certain point, each additional barista contributes less to the overall increase in service speed because the workspace is limited. This scenario exemplifies:

A) Decreasing returns to scale.

B) Diseconomies of scale.

C) Diminishing marginal returns.

D) Negative marginal returns.

7) In the health insurance market, it tends to be the case that individuals looking to purchase health insurance have more information about their own health, lifestyle, and medical history than the insurance company. This is an example of:

A) Asymmetric Information

B) Adverse Selection

C) Moral Hazard

D) Consumer Surplus

8) Alice is willing to pay up to £150 for a concert ticket, but she finds and purchases a ticket for £100. What is Alice's consumer surplus from this purchase?

A) £0

B) £50

C) £100

D) £150

9) Which of the following is likely to be a real-world oligopoly?

A) The market for fast food restaurants in a large city where numerous small chains and independent eateries compete.

B) The global market for commercial aircraft dominated by a few major manufacturers like Boeing and Airbus.

C) The market for organic vegetables at a local farmers' market where multiple small-scale farmers sell similar products.

D) The market for personal computers where consumers have a wide range of brands and configurations to choose from.

10) Which of the following best distinguishes the short run from the long run in economics?

A) In the short run, all factors of production are variable, while in the long run, at least one factor is fixed.

B) In the short run, firms cannot adjust their production levels, while in the long run, firms can adjust their production levels.

C) In the short run, at least one factor of production is fixed, while in the long run, all factors of production are variable.

D) In the short run, firms focus on maximising revenue, while in the long run, firms focus on minimising costs.

Short-answer Questions (please answer both questions)

1) Consider a major airline company that has recently implemented a 15% increase in ticket prices for its domestic flights. After the price change, the company observed a 20% decrease in the quantity of tickets sold.

   a) Based on this information, calculate the price elasticity of demand for the airline's domestic flights. [2 marks]

   Price Elasticity of Demand (PED) = % Change in Quantity Demanded / % Change in Price

   % Change in Quantity Demanded = -20%

   %Change in Price = 15%

   PED = (-20%) / (15%) = -1.33

   b) Explain what the price elasticity of demand calculated above indicates about the responsiveness of customers to changes in ticket prices. [2 marks]

   The price elasticity of demand for the airline's domestic flights is -1.33. Omitting the minus sign, this value is greater than 1, indicating that the demand is elastic. This means customers are relatively responsive to changes in ticket prices. A 1% increase in price leads to a 1.33% decrease in the quantity of tickets sold, showing a greater than proportionate response to price changes.

   c) Discuss at least two potential strategies the airline could use to increase total revenue given this elasticity. [3 marks]

Lowering prices - since the demand is elastic, lowering ticket prices would lead to a proportionately larger increase in the quantity of tickets sold. By reducing prices, the airline could potentially increase its total revenue as the gain from selling more tickets would outweigh the loss from the lower price per ticket.

Enhancing non-price differentiation - the airline could focus on improving service quality, adding more amenities, or offering better loyalty programs. By differentiating its services, the airline might reduce the price sensitivity of its customers, making them more willing to pay higher prices for the added value. This could help in retaining customers even with higher prices and potentially increasing total revenue.

d) Illustrate on a diagram what would happen to the equilibrium price and quantity of ticket prices for domestic flights if technical issues lead to the grounding of specific aircraft models at the same time as peak holiday season arrived. [3 marks]

The technical issues would lead to a leftwards shift in supply and the arrival of peak holiday season would lead to a rightwards shift in demand. The y axis should be labelled P and the x axis labelled Q. The original and new equilibriums should be shown. There would be an increase in the equilibrium price. The number of tickets may rise, fall or stay the same depending on the relative size of each shift.

2) A firm which prints educational textbooks has fixed costs of £1200 and average variable costs (AVC) as shown in the table.

Output (Q) (units)	TFC (£)	AFC (£)	TVC (£)	AVC (£)	TC (£)	AC (£)	MC (£)
0	1200	-	0	-		-	-
	-	-	-	-	-	-	
10				12			-
	-	-	-	-	-	-	

20				11			-
	-	-	-	-	-	-	
30				10			-
	-	-	-	-	-	-	
40				10			-
	-	-	-	-	-	-	
50				12			-
	-	-	-	-	-	-	
60				15			-
	-	-	-	-	-	-	
70				20			-

Output (Q) (units)	TFC (£)	AFC (£)	TVC (£)	AVC (£)	TC (£)	AC (£)	MC (£)
0	1200	-	0	-	1200	-	-
	-	-	-	-	-	-	12
10	1200	120	120	12	1320	132	-
	-	-	-	-	-	-	10
20	1200	60	220	11	1420	71	-
	-	-	-	-	-	-	8
30	1200	40	300	10	1500	50	-
	-	-	-	-	-	-	10
40	1200	30	400	10	1600	40	-
	-	-	-	-	-	-	20
50	1200	24	600	12	1800	36	-
	-	-	-	-	-	-	30
60	1200	20	900	15	2100	35	-
	-	-	-	-	-	-	50
70	1200	17.1	1400	20	2600	37	-

a) Calculate the firms total fixed costs, average fixed costs, total variable costs, total costs, average costs and marginal costs for each level of output and add them to the table. [3 marks]

b) Explain how we know that these are the firm's short run costs. [1 mark]
Both variable and fixed costs are shown. In the long run all costs are variable.

c) Explain the circumstances under which a firm may decide to stop production in the short run. [2 marks]

It may be that there is no output at which the firm can make a profit. In the short-run, fixed costs must be paid even if the firm is producing nothing. Providing that the firm is more than covering its variable costs, it can go some way to paying off its fixed costs and will therefore continue to produce. A firm will shut down if it

cannot cover its variable costs, this will be where the price (AR) is below the AVC curve.

d) A typical firm will have a U-shaped average cost curve.

(i) With the help of a diagram, show why a firm with a steeper average cost curve is more susceptible to any change in demand than a firm with a flatter average cost curve.

[2 marks]

If there is a fall in demand, the firm with a steeper AC curve will experience a significantly faster rise in costs than a firm with a flatter average cost curve. The diagram should show a steep and a flat AC curve and the difference in AC (y axis) when Q (x axis) falls.

If there is an expansion in demand, both firms will increase output, but the firm with the steeper AC curve will see a much faster increase in costs.

(ii) Give two reasons why a firm might have a steep average cost curve.

[2 marks]

1. If a firm has a high ratio of fixed factors to variable factors, it is likely to face a steeper AC curve at lower levels of output.
2. If a firm is relatively inflexible in its use of inputs, it may find that a cut in production means that efficiency falls and average cost rises rapidly. Similarly, if it wishes to expand output, it may find it difficult to do so without incurring considerable extra costs, for example by employing expensive agency staff or hiring expensive machinery. Expanding production might also lead to overuse of machinery, resulting in higher maintenance and operational costs or the firm may face difficulties in effectively managing and coordinating a larger workforce and more complex operations, leading to inefficiencies.

Discussion Questions (please answer only *one* of the following questions)

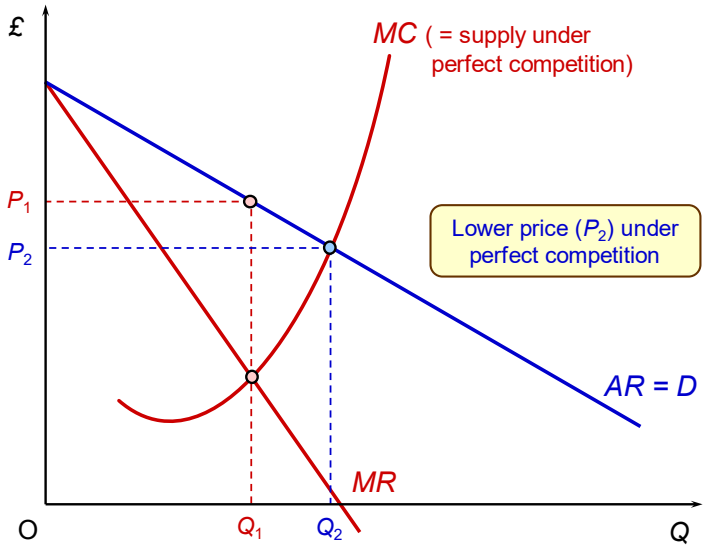
1) Explain why the profit maximising price and the profit maximising level of output of a monopolist will be different to the profit maximising price and profit maximising level of output of an industry operating under perfect competition.

[15 marks]

The monopolist will produce a lower output at a higher price in the short run.

Assume that they both (industry under monopoly and industry under perfect competition) face the same demand curve.
Assume initially that they face the same cost curves.
The monopolist will produce Q1 at a price of P1. This is where MC = MR.
If the same industry were under perfect competition, it would produce Q2 at P2 – a higher output and a lower price.

Equilibrium of industry under perfect competition and monopoly: with the same *MC* curve



For each of the firms in the industry under perfect competition, marginal revenue is the same as price (because firms are price takers).

The firm under perfect competition faces a perfectly elastic demand (AR) curve, which also equals MR. Therefore, producing where MC = MR also means producing where MC = P. When all firms under perfect competition do this, price and quantity in the industry will be given by P2 and Q2.

The monopolist may also produce a lower output at a higher price in the long run. Under perfect competition, freedom of entry eliminates supernormal profit and forces firms to produce at the bottom of their LRAC curve, keeping down long-run prices. A monopolist, however, because of barriers to entry, can continue to earn supernormal profits in the long run and is not forced to operate at the bottom of the AC curve. Ceteris paribus, long-run prices will tend to be higher, and hence output lower, under monopoly.

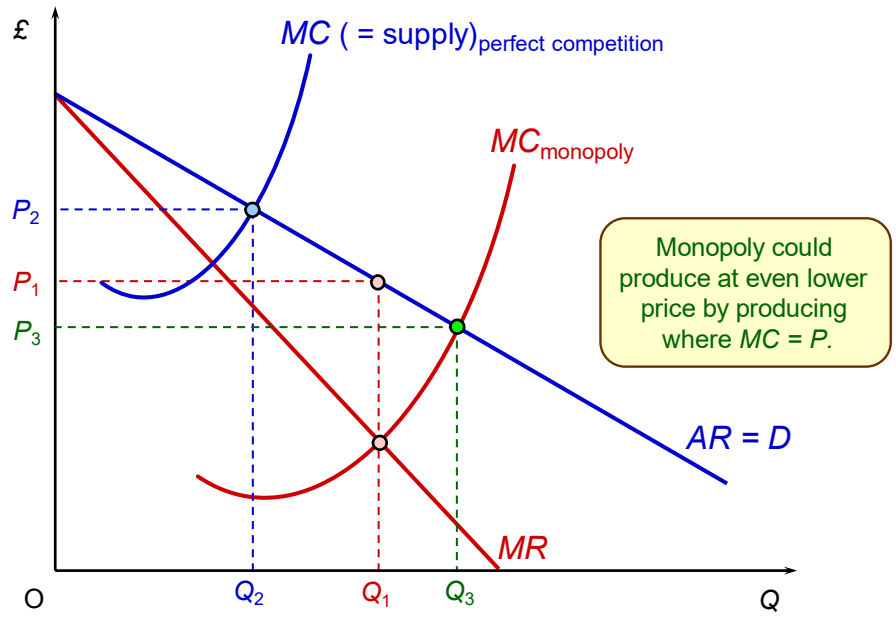
The sheer survival of a firm in the long run under perfect competition requires that it uses the most efficient known technique and develops new techniques wherever possible.

The monopolist, however, sheltered by barriers to entry, can still make large profits even if it is not using the most efficient technique. It has less incentive to be efficient.

On the other hand, the monopoly may be able to achieve substantial economies of scale due to larger plant, centralised administration and the avoidance of unnecessary duplication. If this results in an MC curve substantially below that of the same industry under perfect competition, the monopoly may even produce a higher output at a lower price.

Another reason why a monopolist may operate with lower costs is that it can use part of its supernormal profits for R&D and investment. It may not have the same incentive to become efficient as the perfectly competitive firm that is fighting for survival, but it may have much greater ability.

Equilibrium of industry under perfect competition and monopoly: with different *MC* curves



Although a monopoly faces no competition in the goods market, it may face an alternative form of competition in financial markets. A monopoly, with potentially low costs, which is currently being run inefficiently, is likely to be subject to a takeover bid from another company.

2) Increases in costs of production have had a significant impact on many businesses. Discuss how the costs of production of a firm are influenced by the price and productivity of the factors of production. Distinguish between the short run and long run and use specific examples to illustrate your points. [15 marks]

The factors of production are land, labour, capital and enterprise.
A firm's costs of production will depend on the factors of production it uses.

The greater their physical productivity, the smaller will be the quantity of them that is needed to produce a given level of output, and hence, the lower will be the cost of that input.
Might mention that workers will be more productive if they are more skilled/educated/trained/motivated and have good equipment to use.
There is a direct link between TPP, APP and MPP and the costs of production.
Diminishing returns are important in the short run.
The higher the price of the factors of production, the higher will be the costs of production. In the short tun, some factors are fixed in supply. Therefore, the TC of these inputs are fixed and do not vary with output. The cost of variable factors does vary with output, for e.g. the more is produced, the more raw materials (and potentially labour) are needed and the higher is their cost.
Might need to pay workers overtime at a higher wage if you want to increase production.
Also helpful to bring in high global inflation in recent years – for e.g. firms have had to endure higher oil prices, food prices and wages for example.
In the long run, economies of scale are important – increasing the scale of production leads to lower cost per unit of output.
Diseconomies of scale could also be mentioned.
In the long run, all factors can be varied, so a profit-maximising firm will want to use the least costly combination of factors of production to produce any given output. The least cost combination will be where the extra MPP from the last pound spent on each factor is equal. If they were not equal, it would be possible to reduce cost per unit of output, by using a different combination of factors.
Discussion of opportunity cost.